Ordinal Logistic Regression - RA Treatment Response

Introduction

The Arthritis dataset (from the *vcd* package) is from a double-blind, placebo-controlled trial of auranofin for rheumatoid arthritis. The response variable, *Improved*, is an ordered factor (None < Some < Marked). Predictors include Treatment, Sex, and Age. Because the outcome is ordinal, ordinal logistic regression is appropriate, and *Improved* was confirmed as an ordered factor in R before modelling.

1 Understanding the Model

1.1 Ordinal logistic regression is appropriate

The response variable *Improved* is ordinal (None < Some < Marked), so linear regression is inappropriate and binary logistic regression would lose information. Ordinal logistic regression is suitable as it preserves the ordering and models the probability of higher improvement under the proportional odds assumption, which assumes constant effects across cut-offs.

1.2 Proportional odds model

Formally, for an ordinal response Y with K categories:

$$log\ \frac{P(Y \leq k)}{P(Y > k)} = \alpha_k - \beta_1 X_1 - \beta_2 X_2 - \cdots - \beta_p X_p, \quad k = 1, \ldots, K-1$$

where$\beta_1 \ldots, \beta_p$ are the same across all cutpoints k, and only the intercept α_k changes with each threshold. For K ordered categories, the model requires K − 1 thresholds; since *Improved* has 3 levels, 2 thresholds are needed.

1.3 Proportional odds assumption

The proportional odds assumption says that the effect of each predictor (e.g., Treatment, Sex, Age) is the same across all thresholds of the ordinal outcome. In other words, the log-odds of being treated versus placebo are assumed to be constant, whether predicting None vs. Some/Marked or None/Some vs. Marked. If this assumption is violated, the effect of a predictor differs across thresholds, making the model potentially inappropriate.

2 Exploratory Analysis

2.1 Frequency distribution

Table 7: Overall distribution of Improved

Improved	**n**	**%**
None	42	50.00%
Some	14	16.70%
Marked	28	33.30%

Caption: Frequency and percentage of patients showing None, Some, or Marked improvement.

Table 8: Distribution of Improved by Treatment

Treatment	None	Some	Marked	Total
Placebo	29 (67.4%)	7 (16.3%)	7 (16.3%)	43
Treated	13 (31.7%)	7 (17.1%)	21 (51.2%)	41
Total	42 (50.0%)	14 (16.7%)	28 (33.3%)	84

Caption: Row-wise percentages and counts of patients showing None, Some, or Marked improvement by Treatment group.

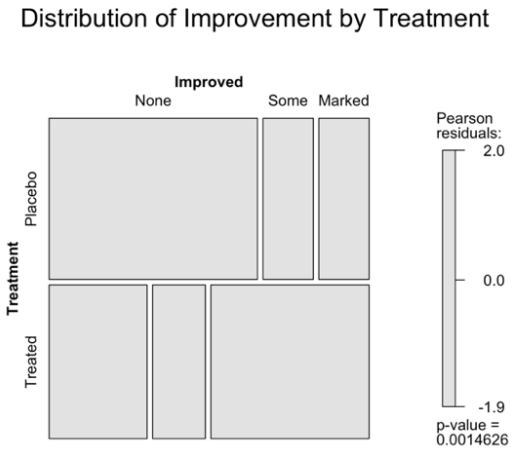
Patients receiving the Treated intervention were more likely to experience Marked improvement (51.2%) compared with those in the Placebo group (16.3%) (see Table 8). Conversely, the Placebo group had a higher proportion of patients showing no improvement (67.4%) relative to the Treated group (31.7%) (see Table 8). Overall, although half of all patients showed no improvement, the distribution of improvement across the treatment groups (see Table 7) suggests a potential effect of the Treated intervention even before formal statistical modelling.

2.2 Summary Statistics for Predictors

The mean age was 53.4 years (SD = 12.8, range 23–74). The Treatment groups were roughly balanced (Placebo 43, 51.2%; Treated 41, 48.8%), while females predominated (59, 70.2%). All cell counts were adequate, with no sparsity concerns.

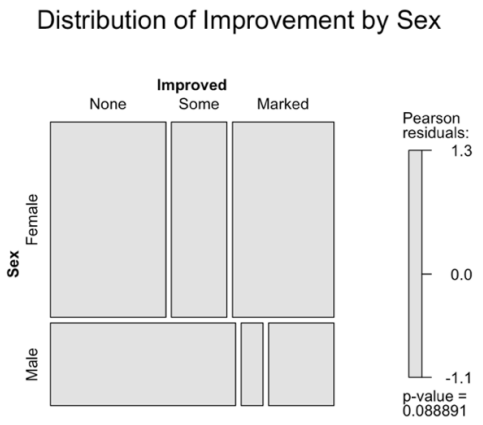
2.3 Distribution of Patient Improvement by Treatment and Sex

Figure 9: Distribution of Improved across Treatment



Caption: It shows the proportion of patients with different improvement levels in Placebo and Treated groups.

Figure 10: Distribution of Improved across Sex

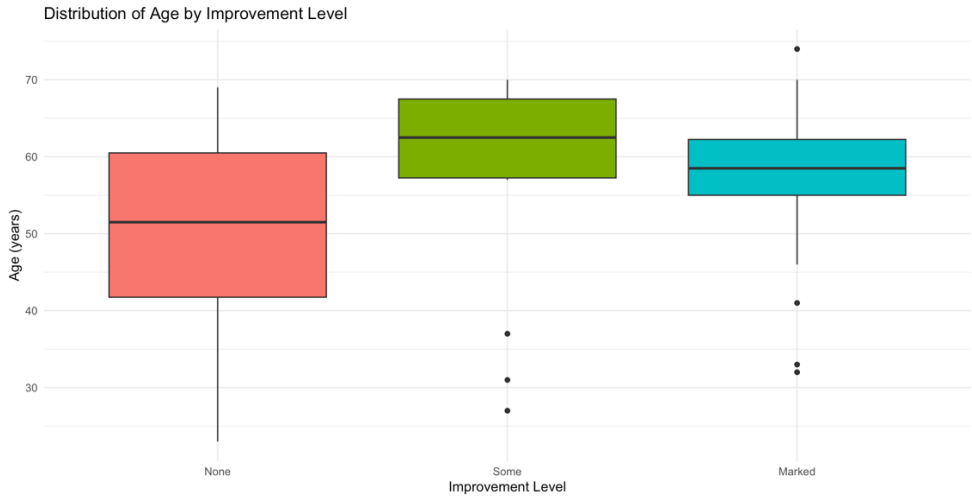


Caption: It shows the distribution of improvement levels among female and male patients.

Mosaic plots were used to show improvement levels across groups. In Figure 9, the Placebo group has the most patients with no improvement, while the Treated group shows more Some and Marked improvement (p = 0.00146), indicating a significant treatment effect.
In Figure 10, improvement is similar between sexes (p = 0.0889), suggesting no association with sex.

2.4 Distribution of Age by level of Improved

Figure 11: Box plot of Age distribution by level of Improved



As shown in Figure 11, the median age is highest in the Some group and lowest in None, with None showing the widest range. Older individuals tend to show Some or Marked improvement, suggesting a tentative association between age and outcome.

3 Fitting and Interpreting the Model

3.1 Proportional odds model

A proportional odds model was fitted with Improved as the response and Treatment, Sex, and Age as predictors, with reference levels set to Placebo and Female. Results indicate Treated

participants are more likely to improve (β = 1.745), males less likely than females (β = -1.252), and older age slightly increases improvement (β = 0.038). Thresholds for None|Some and Some|Marked are 2.532 and 3.431, respectively; residual deviance is 145.46 and AIC 155.46.

3.2 Fitted logit equation

The fitted cumulative logit equations are:

Threshold 1: None vs Some/Marked

$$log\frac{P(Y \leq None)}{P(Y > None)} = 2.532 - 1.745 \text{ Treatment Treated} + 1.252 \text{ SexMale} - 0.038Age$$

Threshold 2: None/Some vs Marked

$$log\frac{P(Y \leq Some)}{P(Y > Some)} = 3.431 - 1.745 \text{ Treatment Treated} + 1.252 \text{ SexMale} - 0.038Age$$

Here, the thresholds 2.532 and 3.431 separate None vs Some/Marked and None/Some vs Marked, respectively, while the coefficients represent the effects of Treatment, Sex, and Age on the cumulative odds of being at or below each improvement level.

3.3 Clinical interpretation for odds ratios

The coefficients from the proportional odds model were converted to odds ratios (ORs) using the formula: $OR = e^{\beta}$. Patients who received the treatment have 5.7 times higher odds of being in a better improvement category, indicating a strong treatment effect. Male patients have 71% lower odds (1 − 0.29) of being in a higher improvement category compared to female patients, indicating females tend to improve more than males in this dataset. Each additional year of age increases the odds of being in a higher improvement category by 4%, indicating older patients show a slight tendency to improve more than younger patients.

3.4 Proportional Odds Model Results

Treatment is the strongest predictor (β = 1.745, p = 0.0004), with Treated participants more likely to improve. Sex also matters, with males less likely to improve than females (β = -1.252, p = 0.0246). Age shows a modest positive effect (β = 0.038, p = 0.0415). Overall, treatment drives improvement, while sex and age contribute modestly.

4 Checking the Proportional Odds Assumption

4.1 Proportional Odds Assumption Test

Table 9: Brant Test for Proportional Odds Assumption

Predictor	**χ^2**	**df**	**p-value**
Treatment (Treated)	0	1	0.99
Sex (Male)	1.77	1	0.18
Age	1.02	1	0.31
Omnibus	3.03	3	0.39

H_0: Predictor effects are the same across all outcome thresholds (proportional odds holds)
H_1: Predictor effects differ across thresholds (proportional odds is violated)

As shown in Table 9, all p-values are greater than 0.05, so we fail to reject H_0. There is no strong evidence that the proportional odds assumption is violated, and the effect of each predictor appears reasonably consistent across thresholds.

4.2 Implications of a Violation of the Proportional Odds Assumption

A violation of proportional odds means a predictor's effect varies across thresholds. In this trial, if the assumption were violated, it would mean a predictor's effect differs across improvement levels. Clinically, the main conclusions (e.g., Treatment is effective) would likely remain, but reporting would be more complex, requiring separate effects for each threshold.

4.3 Likelihood Ratio Test Comparing Fitted and Null Models

Likelihood Ratio Test:
H_0: The predictors (Treatment, Sex, Age) do not improve model fit (all coefficients = 0)
H_1: At least one predictor improves model fit (at least one coefficient ≠ 0)
The likelihood ratio test statistic is 24.46(df = 3), and the p-value is 2.0×10^{-5}, much smaller than 0.05. Therefore, we reject H_0. This indicates that the model including Treatment, Sex, and Age fits the data significantly better than the intercept-only model. Clinically, Treatment, Sex, and Age all contribute to predicting patient improvement, and including them improves the prediction compared to assuming all patients have the same probability of improvement. Deviance comparison: null deviance = 169.92, residual deviance = 145.46. The difference is 24.46, which represents representing the reduction in deviance explained by adding the predictors.

5 Interpretation and Limitations

5.1 Key Findings

Treatment, Sex, and Age were significantly associated with improvement. Receiving treatment increased the odds of higher improvement, males were less likely to improve, and older participants showed slightly better improvement. Overall, treatment is the strongest driver of improvement, with sex and age contributing moderately.

5.2 Caution on causation

While the odds ratio for Treatment is greater than 1, indicating that treated patients are more likely to improve, we should be careful with the word "causes." A randomised controlled trial allows us to infer a causal effect only if the study is properly randomised, blinded, and free from bias or confounding. In this analysis, the model estimates associations between predictors and improvement, but strictly speaking, the statistical output alone shows association, not definitive causation. The causal interpretation is supported by the trial design, not by the regression coefficients themselves.

5.3 Sample Size Consideration

A relatively small sample size (n = 84) can lead to less precise estimates of coefficients, reflected in larger standard errors and wider confidence intervals, making it harder to detect

true effects. In a follow-up study, a researcher might increase the sample size to improve the reliability and stability of model estimates and allow more accurate estimation of treatment effects.